the upper row of Figure 11.5, and then doing the same for the final Calabi-Yau shape from the lower row of Figure 11.5. If the mirror relationship is not shattered by the tear in the upper Calabi-Yau, then these two final Calabi-Yau shapes should yield identical physics, just like the two initial Calabi-Yau shapes from which they evolved. (This way of phrasing the problem avoids doing any of the very difficult calculations involving the upper Calabi-Yau shape just when it tears.) It turns out, though, that calculating the physics associated with the final Calabi-Yau shape in the upper row is pretty straightforward. The real difficulty in carrying out this program lies in first figuring out the precise shape of the final Calabi-Yau space in the lower row of Figure 11.5—the putative mirror of the upper Calabi-Yau—and then in extracting the associated physics.

A procedure for accomplishing the second task—extracting the physical features of the final Calabi-Yau space in the lower row, once its shape was precisely known—had been worked out a few years earlier by Candelas. His approach, however, was calculationally intensive and we realized that it would require a clever computer program to carry it out in our explicit example. Aspinwall, who in addition to being a renowned physicist is a crackerjack programmer, took on this task. Morrison and I set out to accomplish the first task, namely, to identify the precise shape of the candidate mirror Calabi-Yau space.

It was here that we felt Batyrev's work could provide us some important clues. Once again, though, the cultural divide between mathematics and physics—in this case, between Morrison and me—started to impede progress. We needed to join the power of the two fields to find the mathematical form of the lower Calabi-Yau shapes that should correspond to the same physical universe as the upper Calabi-Yau shapes, if flop tears are within nature's repertoire. But neither of us was sufficiently conversant in the other's language to see clear to reaching this end. It became obvious to both of us that we needed to bite the bullet: Each of us needed to take a crash course in the other's field of expertise. And so, we decided to spend our days pushing forward as best we could on the calculation, while spending evenings being both professor and student in a class of one: I would lecture to Morrison for an hour or two on the relevant physics; he